Artem Lavrov

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Education

Brandeis University — *Waltham, MA*
B.S. in Computer Science, Minor in Mathematics — *May 2026*
- GPA: 3.89, Summa Cum Laude, High Honors, Dean's List
- **Relevant Coursework:** Operating Systems, Probability, Distributed Systems, Theory of Computation, Intro to Computer Networking, Algorithms, Database Management Systems

Skills

Programming Languages: Rust, Java, Python, TypeScript, JavaScript, C, Go, C#
Operating Systems: Linux (Ubuntu, Arch, Debian, NixOS)
Tools & Technologies: Git, GitHub, Docker, Azure Cloud, REST APIs, Maven, Selenium, SQL
AI Tools: Claude Code, OpenCode

Technical Experience

Researcher (Senior Thesis) - Tectonic + — *Waltham, MA*
Smart and Scalable Data Systems Lab — *Dec 2025 - May 2026*
- Extended an inherited workload generator into a full Rust benchmarking suite, building its execution engine and statistics collection (per-operation p95/p99 latency, throughput, success and total operation counts), profiling with flamegraphs and Valgrind to keep harness overhead off the measurement path.
- Measured 1.3–2.4x higher throughput and ∼86% lower peak memory than YCSB, the industry-standard key-value benchmark, across all six of its core workloads with the same configuration of RocksDB and running on identically configured Chameleon Cloud bare-metal nodes.
- Built an algorithm that reads each workload section's declared operation mix and composes the simplest sufficient key-tracking structure from hash sets, bloom filters, and sorted or unsorted vectors.
- Designed a storage-engine-agnostic interface for key-value stores, implemented for four embedded, networked, and distributed databases (RocksDB, Redis, Cassandra, ScyllaDB), using static dispatch and per-database feature flags so builds pull only the drivers they need.

Teaching Assistant — *Waltham, MA*
Brandeis University — *Jan 2025 - Dec 2025*
- **Software Engineering (2 sections):** Reviewed code in person with 4 student project teams over their PR diffs, evaluating modularity, code reuse, and function decomposition, and testing their REST API submissions with curl.
- Supported 50+ students through weekly recitations, fielding questions on Git, Maven, and general implementation problems in their projects.
- **Computer Security (2 sections):** Built a Python grading tool that auto-resolved roughly 70% of submissions and routed the rest for manual review, with resumable state and CSV export for bulk grade upload.

Research Assistant — *Waltham, MA*
Brandeis University — *Dec 2024 - May 2025*
- Solo-implemented a semester-long project scoped for 3–4 person teams, building a Java client/server prototype with REST endpoints against the instructor's API.
- Identified specification and API ambiguities where the described requirements were open-ended but the expected solution was not, driving revisions to the project documentation before the term began.

Customer Relationship Management Intern — *Delaware, OH*
Greif — *May 2024 - Jul 2024*
- Implemented EU sanctions-driven data separation across a heavily customized SugarCRM instance, updating roles for hundreds of users so Russian and non-Russian teams could not access each other's data.
- Built custom SQL reports to verify no account retained access breaching the sanctions policy, several of which the CRM team kept as standing reports after the migration.

Project Experience

Dominion Engine & Simulator *TypeScript, React, Zustand, Hono, Bun, WebSockets, GitHub Actions*
Personal Project — *May 2026 - Present*
- Built an authoritative multiplayer server holding all hidden state, sending each player a filtered view over WebSockets so opponents' information never reaches the client.
- Engineered a scheduler resolving game effects as a queue of resumable steps, sequencing nested effects and interrupts around steps that pause for player input.
- Built the core engine and API a collaborator implemented all 33 base-set card effects against.
- Implemented LLM players on the engine API, able to select only from engine-supplied legal moves.
- Built a 108-test suite with Claude Code covering all 33 base-set card effects, run in CI to block merges on failure.
- Runs ongoing 4-player games on a self-hosted server, with reconnect that restores state and re-issues pending decisions.